

QuantFundAgent

A quick catch-up on my thesis project

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The purpose of this document is to give an overview of the most important changes since our last talk in the first section and to discuss the comparisons of some researched alpha factors as well one first walk forward backtest in the subsequent sections.

1 Important changes since last meeting

- Downloaded and preprocessed LOBSTER data into the standard 10-second bin format for ten ETFs across different sectors/assets: SPY, QQQ, IEF, FXI, FXE, CORN, CPER, GLD, SHY, UFO; sample period: 2024-01-01 to 2026-06-01.
- Modularised the agent framework around the active configuration. Factor and strategy artefacts are now stored separately under `data/workspaces`, which keeps universes and LLM model choices separated and makes results easier to compare.
- Compared three factor sets on the 10-second LOBSTER data: factors generated by GPT-4o-mini, factors generated by GPT-5.4-mini, and the seed factors; results and methodology are discussed below.
- Added a rolling factor-comparison pipeline with monthly in-sample/out-of-sample windows per ticker, including aggregated feature-importance and performance outputs.
- Created a point-in-time S&P 500 membership list from historical constituent data and Wikipedia change logs; delisted tickers are tracked, although their price history is not supported by yfinance.
- Improved the Architect/backtesting logic, including clearer position construction, execution-cost handling, and support for injecting pre-generated factor sets into walk-forward runs.
- Backtested a walk-forward run of the full agentic framework on a static S&P 100 list from 2016; results are discussed below.
- added after current tests described below: factor researcher adds forecast horizon to every factor when creating it

2 Factor comparison

2.1 methodology

The factor comparison is designed as a controlled comparison of the factor libraries, not as a full fund backtest. I compare three factor sets on the same 10-second LOBSTER panel: 66 factors generated by 68 GPT-4o-mini, factors generated by GPT-5.4-mini, and the seed factor library resulting from [1] consisting of 88 factors. The cost for computing the LLM-generated factors were 0.32\$ and 2.00\$ respectively. All sets are evaluated on the same ETF universe and data period, so the intended difference between runs is the source of the factor library.

Because the amount of data is so large, the script builds rolling monthly windows with two in-sample months followed by one out-of-sample month. Each window and ETF is evaluated separately. This avoids fitting on the whole sample at once and gives a cleaner view of whether factor relevance is stable over time.

Within each window the comparison evaluates single-factor information coefficients, factor redundancy and diversity, and a machine-learning combination of the factor signals. The machine-learning track is trained only on the in-sample period and evaluated on the following out-of-sample month using a vectorised backtest. The downstream agentic fund is not run here; the aim is to isolate factor-set quality before adding strategy selection and portfolio-management decisions.

The first set of metrics is calculated at the single-factor level. For each usable factor, the factor signal is compared with the ETF's own future return using an information coefficient. In the current setup this is a time-series IC per underlying. The comparison also records IC stability through an IC information ratio and the sign hit rate. At the factor-library level, I additionally calculate pairwise factor correlations, the effective number of independent factors, redundancy clusters, and model-based feature importance. The independence measure is based on the eigenvalues of the factor correlation matrix: if all factors are nearly uncorrelated, the effective number is close to the raw number of factors, while highly similar factors reduce it towards one. Redundancy is measured in two related ways. First, factors are grouped into clusters when their absolute correlation is above the chosen threshold, so a large cluster indicates several factors are capturing a similar signal. Second, the redundancy score is calculated from the gap between the raw number of factors and the effective number of independent factors. Feature importance is estimated by fitting simple predictive models, mainly lasso and gradient boosting, from the factor signals to future returns and then measuring which factor columns receive the largest absolute coefficients or model importance values.

The machine-learning part tries to combine the factors as input features to create signals with stronger predictive power. For one ETF and one rolling window, each row is one timestamp, the columns are the standardised factor signals, and the target is the ETF's future return over the selected prediction horizon. The model is fitted on the two in-sample months only and then used to predict a combined return signal for the out-of-sample month.

In addition to the individual models, I also evaluate a simple ensemble signal. This is not fitted as a separate model. Instead, each successfully fitted model first produces its own predicted return signal; these model signals are divided by their respective standard deviations to create proper comparability. The ensemble therefore measures whether combining several model-based factor combinations gives a more stable signal than relying on a single model specification.

This predicted return signal is then converted into positions using a simple rule. The signal is standardised over time for the ETF; by default the backtest goes long when the standardised signal is above one standard deviation of its historical mean, short when it is below the one standard deviation of it, and otherwise stays flat. Positions are implemented as a staggered holding-period book: each signal contributes only one tranche of exposure, which remains active for the holding period, and overlapping tranches are averaged into the live position before P&L is calculated from one-bar forward returns. Here I use 6 bar forecasts, so a long signal will result in the following six bars trading $\frac{1}{6}$ of the ETF. Another long signal right after this would lead to the five following bars trading $\frac{2}{6}$ and the sixth following again $\frac{1}{6}$ of the ETF. This version assumes the (probably unrealistic) ability to immediately trade after one bars close and on the next bars open. This gives the realised P&L series from which performance metrics are calculated.

The main machine-learning performance metrics are in-sample and out-of-sample IC, Sharpe ratio, annualised return, maximum drawdown, and hit rate. I mainly use the out-of-sample values to judge whether the factor set contains useful predictive information, while the difference between in-sample and out-of-sample Sharpe gives a basic indication of overfitting.

2.2 results

Before evaluating the results, it is critical to highlight that the standard formula for annualizing the Sharpe ratio,

$$Sharpe_{Annualized} = \frac{E[R_t]}{\sigma(R_t)} \times \sqrt{N} \quad (1)$$

includes a massive scaling factor of $\sqrt{589,680} \approx 767.9$ when working with 10-second sampling intervals. This time-aggregation multiplier drastically inflates the absolute value of the annualized metric. Consequently, the reported Sharpe ratios should be interpreted strictly as an informational tool to evaluate the relative performance across different factor sets and underlyings, rather than as a realistic claim about future out-of-sample absolute performance. This relative validity is supported by Figure 2a, which demonstrates a strong empirical correlation between the out-of-sample information coefficient (IC) and the corresponding out-of-sample Sharpe ratio.

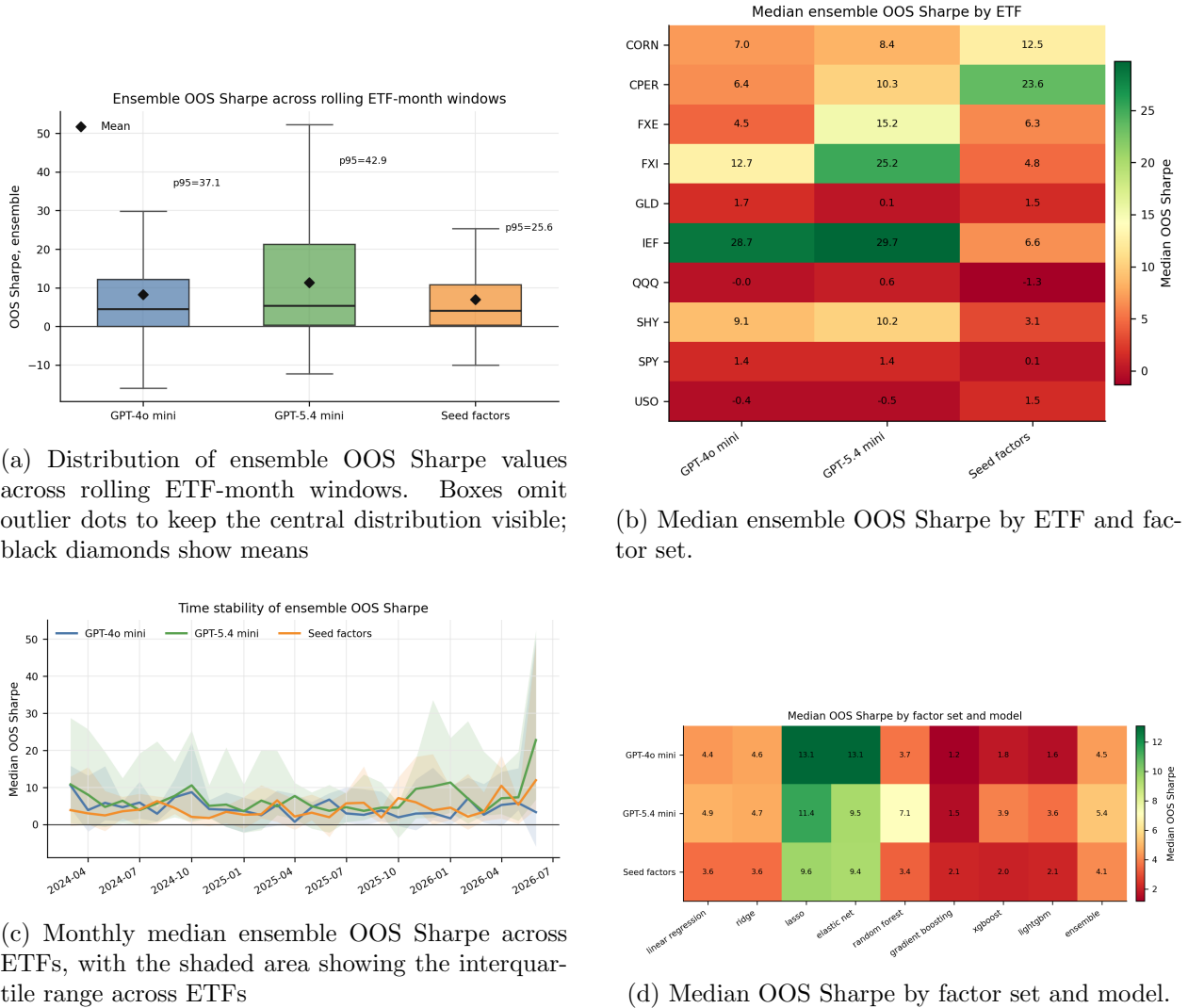
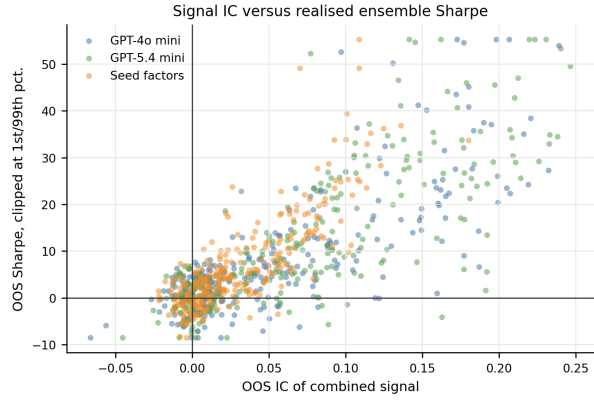


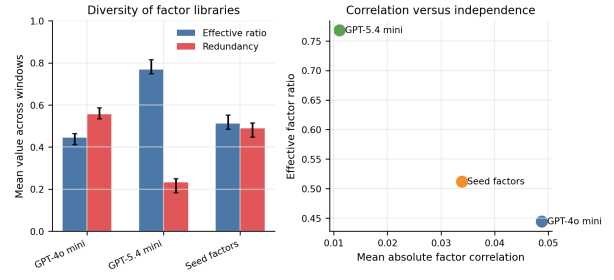
Figure 1: OOS Sharpe Analysis of different factor sets, models and ML models

Observations:

- The factors found by gpt-5.4-mini show significantly greater indepenence than those found by gpt-4o-mini and the seed factors (2b)
- Trading performance resulting from 5.4-mini factors is the best, though it is not as significant as the ability to find independent factors. Especially 5.4-mini has more positive outliers 1a.



(a) Relationship between OOS information coefficient and realised ensemble Sharpe. The Sharpe axis is percentile-clipped for readability.



(b) Factor-library diversity. The effective ratio is the eigenvalue-based effective number of independent factors divided by the raw number of factors; redundancy is one minus this ratio. Error bars show the interquartile range across windows.

Figure 2: Analysis of relation between out-of-sample IC and sharpe reatio in a) and independence of factors in b)

- Trading performance also depends heavily on the underlying being traded with SPY and QQQ having the worst performance and IEF having the best [1b](#)
- Across all factor sets, the lasso and elastic-net models produce the strongest OOS prediction and trading metrics. Since these regularised linear models shrink many coefficients to zero, this suggests that the useful predictive information is concentrated in a relatively small subset of factors rather than being spread evenly across the full library.
- General tendency of better LLM -> better results, overall promising results
- More detailed results can be found in [data/comparisons/rolling_20260624_001308](#)

3 Walk-forward backtest

3.1 methodology

The walk-forward backtest tests the full agentic framework in a through-time setting. It uses daily yfinance data for a static S&P 100 universe from 2017-01-01 to 2026-06-01. Since the universe is fixed over the whole period, this should be read as a practical baseline rather than a survivorship-bias-free historical index reconstruction.

The framework advances on a monthly grid: at each due date, a research step, a strategy-construction step, and a portfolio-management step can occur. Instead of running fresh LLM factor research during the backtest, the research step injects two pre-generated factors per month from the sp100-5.4-mini and sp100-4o-mini preruns (116 and 101 factors respectively), while the seed factors remain available throughout the whole backtest. This makes the run more reproducible and limits token cost.

At each strategy meeting, the framework runs the same Selector–Architect–Statistician pipeline that is used in the normal agent workflow, but with the market data truncated at the current cutoff date. The Selector first forms a trading hypothesis and selects factors from the factor catalogue available at that point in the backtest. In this run, new factors are not generated by the LLM during the backtest. Instead, two pre-generated factors are injected at each research step from the preruns, while the seed factors remain available throughout.

The Architect then turns the selected factors into a candidate strategy. For an ML-based strategy, the factor values are arranged as a supervised learning problem where each sample corresponds

to one timestamp and ticker, the features are the normalised factor signals, and the target is the future return over the chosen prediction horizon. The available pre-cutoff data is split chronologically: the 20% most recent data is reserved for out-of-sample testing by the Statistician, while the earlier part is used by the Architect. (Important to note: Always all available data is used for training the model. Should it better be a fixed amount like for example always last 12 months with 9 IS and 3 OOS?) Inside the Architects training window (the 80% in sample), the model is first fit on a train sub-window and checked on a validation sub-window again using an 80/20 IS/OOS split. After this result the architect can either refit or keep the model if he is satisfied. In the current setup an architect can run a maximum of 3 iterations. After these he has to choose one of the models or discard the approach. After selecting one model, the final estimator is refit on the full Architect in-sample window and stored as a model artefact.

Linear models, regularised linear models, random forests, and boosting models are all treated as regressors from factor signals to forward returns. Their output is a predicted return signal for each stock and timestamp. This signal is then passed to the same position-construction layer as other strategies, so the later backtest evaluates whether the predicted signal leads to useful positions after trading rules and costs are applied.

For the S&P 100 walk-forward run, strategy-level positions are constructed cross-sectionally. Each strategy ranks stocks against the current universe at a given date, keeps the strongest signals (the amount of underlyings that are kept differs from strategy to strategy), and rescales the resulting long-short book. The final fund portfolio is not forced to remain fully invested: strategy books are first weighted by the portfolio manager, then netted across strategies and passed through fund-level position constraints. The whole book is essentially a portfolio of strategies that go long the underlying with the relative strongest conviction of performing an upwards move and short the ones with the relative highest conviction of falling, not trading the remaining underlyings or those where strategies produce signals in different directions. This is still quite restrictive and an extension could be to give the portfolio manager agent full freedom in how to use the signals of the different strategies and models to combine them into a portfolio.

During each live window, all active strategies are recomputed on unseen data using their stored model artefacts and current factor signals. The PM allocates capital across the approved strategy book; in this run I use a single balanced PM rather than the committee, partly to reduce token cost. The execution layer then combines all strategy-level target positions into one netted fund book, so offsetting positions in the same stock cancel before trading costs are charged. The book is constrained by the maximum number of names, maximum weight per name, and gross leverage cap of 100 %. Returns are calculated after commission and a synthetic spread cost: because daily yfinance data has no bid-ask spread field, I use a 2 bps fallback quoted spread and charge the half-spread component plus commission on turnover.

3.2 results

The factor research preruns using gpt-4o-mini and gpt-5.4-mini cost 0.6\$ and 3.4\$ respectively. The forward walking backtest used gpt-4o-mini for all its agent and resulted in total cost of 1.31\$.

The most important results benchmarked against OEX (the S&P 100 price return) are shown here. More detailed analytics are stored in data/backtests/sp100_inject.

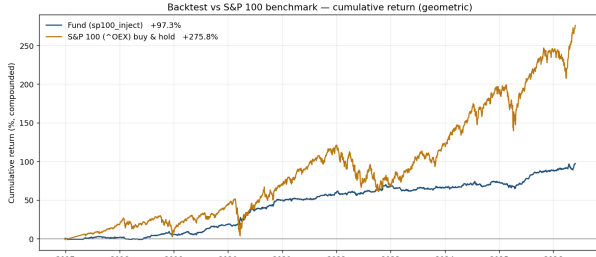
- Absolute performance is below the benchmark, but the fund has much lower realised risk. Annualised volatility is 5.4% versus 19.1% for the S&P 100, and maximum drawdown is about -6.1% versus -31.5%. This leads to a higher Sharpe ratio, 1.40 compared with 0.85 for the benchmark.
- The strategy behaves like a low-volatility, risk-controlled allocation. Average gross exposure is 81.6%, with a median of 80.6%, so the fund is not always fully invested.
- Transaction costs are material. Total cost over the run is about 8.0% of capital, with average

daily turnover of 23.0%.

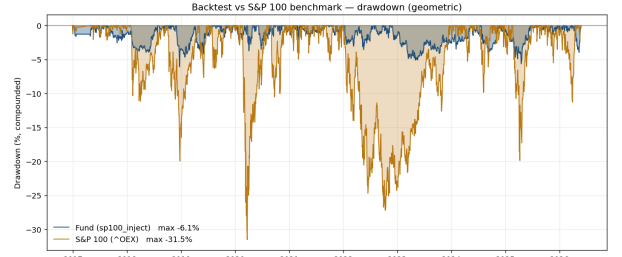
- The portfolio manager is selective over time. Although 50 strategies are approved, only 27 are actually deployed, and by the end of the run only a small number remain live. Many strategies are retired after weak live performance, showing that the PM layer is actively filtering the strategy pool rather than holding all approved strategies permanently.

Table 1: Geometric (compounded) performance of the fund versus the S&P 100 benchmark. NAV is built as $\prod_t(1 + r_t)$; Sharpe and Sortino ratios use $r_f = 0$ and 252 trading days per year.

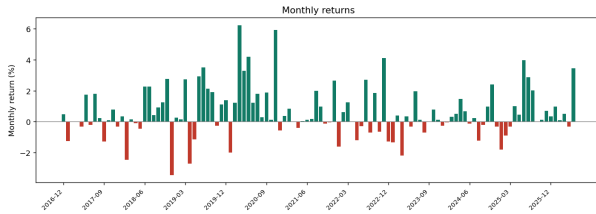
Metric	Fund (sp100_inject)	S&P 100 (^OEX)
Total return (compounded)	+97.29%	+ 275.84%
CAGR (annualised)	+7.71%	+ 15.57%
Annualised volatility	5.40%	19.14%
Sharpe ratio	1.40	0.85
Sortino ratio	2.29	1.21
Max drawdown	− 6.08%	−31.53%
Max DD duration (days)	360	491
Calmar ratio	1.27	0.49
Hit rate (% up days)	47.79%	55.42%
Best day	+2.90%	+ 10.20%
Worst day	− 1.71%	−11.57%
Final NAV (per \$1)	\$1.97	\$3.76



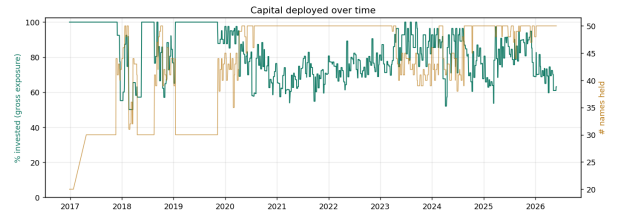
(a) Cumulative Return strategy vs. Benchmark



(b) Drawdown Strategy vs. Benchmark



(c) Strategy monthly returns.



(d) Strategy investment ratio.

Figure 3: Forward-walking backtest results

References

- [1] Zura Kakushadze. 101 formulaic alphas. *Wilmott*, 2016(84):72–80, 2016.